

Surveys (TOR 3)

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Dredge surveys

Sea scallop dredge surveys were conducted by the Northeast Fisheries Science Center (NEFSC) in 1975 and annually after 1977 to measure abundance and size composition of sea scallops in the Georges Bank and Mid-Atlantic regions (Figures 3.1 and 3.2). Means and standard errors were calculated using the standard methods for stratified random surveys (Cochran 1977, Serchuk and Wigley 1989, Smith 1997).

The 1975-1978 surveys used a 3.08 m (10') unlined New Bedford scallop dredge with 54 mm rings. A 2.44 m New Bedford survey dredge with 54 mm rings and a 38 mm plastic liner has been used since 1979. Based on comparisons between camera and dredge data, scallops greater than 40 mm are considered fully selected by the lined survey dredge gear (NEFSC 2007). The northern edge of Georges Bank was not covered by the NEFSC survey until 1982. Data from the Canadian scallop survey during 1979-1981, which used the same gear as the NEFSC survey, was used to evaluate the northern edge in those years (NEFSC 2010). The Canadian and U.S. surveys had comparable catches (Serchuk and Wigley 1986).

The R/V *Albatross IV* was used for all NEFSC scallop surveys from 1975-2007, except during 1990-1993, when the R/V *Oregon II* was used instead. Surveys by the R/V *Albatross IV* during 1989 and 1999 were incomplete on Georges Bank. In 1989, the R/V *Oregon* and R/V *Chapman* were used to sample the Great South Channel and a section of the southeast part of Georges Bank. Serchuk and Wigley (1989) did not find significant differences in catch rates between the R/V *Albatross IV*, R/V *Oregon II* and R/V *Chapman*. The F/V *Tradition* was used to complete the 1999 survey on Georges Bank. NEFSC (2001) found no statistically significant differences in catch rates between the F/V *Tradition* and R/V *Albatross IV* from 21 comparison stations after adjustments were made for tow path length; comparison between the R/V *Albatross IV* and other commercial vessels similarly showed no significant differences (NEFSC 2010). Therefore, survey dredge tows from these other vessels were used without adjustment except for normalizing for tow distance as discussed below.

During 2008-2023, the NEFSC scallop survey was conducted on the R/V *Hugh Sharp*. Direct and indirect comparisons between the catches by the R/V *Hugh Sharp*, R/V *Albatross IV* and commercial vessels towing the lined survey dredge were not significantly different (NEFSC 2010). However, average catches were slightly greater (~5%) on the R/V *Hugh Sharp*. Comparison of tow distance data from dredge sensor data indicate that tow lengths from the R/V *Hugh Sharp* were about 8% longer on average than those on the R/V *Albatross IV* or commercial vessels, suggesting that the slight increase in catch on the *Sharp* is due to its longer tow distance.

As in NEFSC (2014) and NEFSC (2018), each tow was normalized to a tow length of 1 nm. Because dredge sensor data are only available for a subset of the tows, regression

equations were developed based on tows where the sensor data is available to predict tow distance using nominal tow distance and depth as predictors. Nominal tow distance is the nominal tow time, i.e., the time elapsed after the winch is locked at the beginning of the tow to the time when haul back begins, times the mean vessel speed between these times. Separate relationships were developed for the R/V *Albatross IV* (which was assumed to also apply to the other vessels used from 1989-1999), and the R/V *Hugh Sharp*:

$$L = 0.0864 + 0.000444 D + 0.972 N \text{ (R/V Albatross IV)}$$

$$L = -0.0388 + 0.001484 D + 1.061 N \text{ (R/V Hugh Sharp)}$$

where tow length L , and nominal tow length N is in nautical miles and depth D is in meters.

Rock excluder chains have been used on NEFSC sea scallop survey dredges since 2004 in certain hard bottom strata to enhance safety at sea and increase reliability (NEFSC 2004). Based on paired tow trials with and without excluders, the best overall estimate was that rock chains increased survey catches on hard grounds by a factor of 1.31 (CV 0.2). To accommodate rock chain effects in hard bottom areas, survey data collected prior to 2004 from hard bottom areas in the Great South Channel (strata 49-52) and the northern edge (inside Closed Area II) were multiplied by 1.31 when calculating stratified means. Variance calculations in these strata include a term to account for the uncertainty in the adjustment factor (NEFSC 2007).

Relatively high abundance of sea scallops in closed areas makes it necessary to further post-stratify survey data by splitting NEFSC shellfish strata that cross open/closed area boundaries (NEFSC 1999). After re-stratification, the new strata set were combined into open, closed or other areas as required for assessment and management purposes.

The Virginia Institute of Marine Science (VIMS) conducted intensive dredge surveys of selected regions on commercial vessels since 1999, and region-wide dredge surveys of the Mid-Atlantic Bight since 2014. Larger portions of Georges Bank have been surveyed by VIMS since 2016. This includes the Nantucket Lightship, the Southern Flank, Closed Area II, and Closed Area I. In 2020 and 2023 VIMS surveyed the entire resource in the Mid-Atlantic and Georges Bank (except the northern edge of Georges Bank in 2020) because there was no NEFSC surveys in those years. Between 1999 and 2014, the VIMS surveys used partially randomized grid designs, but since then, they have used a random stratified design as in the NEFSC dredge surveys. The VIMS surveys are conducted on commercial scallop vessels using two dredges; the NEFSC lined survey dredge is deployed on one side while a commercial dredge is used on the other side.

All VIMS data for fully covered strata (original or post-stratified) were treated in the same way as NEFSC tows. The partially randomized grid design was treated as random when calculating variances. This likely slightly overstates the true sample variance.

In some years, a few strata, typically those of relatively low density, were unsurveyed. In these cases, densities in these strata were imputed using Generalized Additive Model (GAM) models. The models were fit to estimate trends in average catch rates over time for individual survey strata within subregions. Length composition data for such strata was

estimated as the stratified mean length composition for the surveyed strata in the same subregion.

Capture efficiency of the survey dredge has been estimated by comparing dredge catches to densities observed by the Habcam system towed at the same location (Miller et al., 2018). The best estimates of dredge efficiency were 0.4 on sand substrates, and 0.27 in areas with rougher gravel/cobble substrates. These, together with estimates of tow path length and stock area (see above), were used to expand mean catch per tow and estimate stock size in absolute terms. For these purposes, the Great South Channel north of 40.833 N, and the northern portion of Closed Area II are considered to have gravel/cobble bottom, and other areas, including all of the Mid-Atlantic are assumed to be predominantly sand and are expanded based on a survey dredge efficiency of 0.4. Beginning in 2016, a dredge efficiency estimate of 0.13 was applied to high-density scallop beds to account for reduced dredge efficiency. The reduced capture efficiency is observed when scallop density is ≥ 2 scallops m^{-2} . This value was estimated from paired opportunistic and planned dredge/Habcam tows in 2016-2017 in SARC 65 (see Examination of dredge efficiency using paired tows under TOR 2, NEFSC 2018). All dredge data from 2015-2019 in the Nantucket Lightship South Deep Scallop Area Management Simulator (SAMS) Area was scaled to obtain absolute biomass with the reduced catch efficiency value. In 2021, the reduced catch efficiency value was applied to select stations in the Nantucket Lightship South Deep SAMS Area where dredge stations identified as having densities of ≥ 2 scallops m^{-2} from NEFSC Habcam data were adjusted with the reduced value.

Dredge survey results

Biomass and abundance trends for the dredge survey are presented in Table 3.1 and Figure 3.2. Based on dredge survey estimates, biomass and abundance on Georges Bank were generally low until around 1995. Very large increases were observed during 1995-2000 after implementation of closures and effort reduction measures, but have been slowly declining in recent years. In the Mid-Atlantic Bight, dredge abundance and biomass indices were at low levels during 1979-1997, and then increased rapidly during 1998-2003 due to area closures, reduced fishing mortality, changes in fishery selectivity, and strong recruitment. Biomass was relatively stable during 2003-2008, but then declined, in part due to poor recruitment and fishing down of rotational areas. Mid-Atlantic biomass and abundance has been declining rapidly since about 2018.

Drop camera survey

Region-wide drop camera surveys have been conducted by the School for Marine Science and Technology (SMAST), University of Massachusetts Dartmouth since 2003 (Figure 3.3, Stokesbury et al. 2004, Bethoney et al. 2016, Bethoney and Stokesbury 2018). Each survey station consists of a cluster of four nearby drops, and stations are placed on a grid generally either 1.5 nmi (2.78 km) or 3 nmi (5.56 km) apart, although in some areas they are closer together. There were several cameras on the camera pyramid, but the survey index through 2015 is based on the “large” camera, which was a standard definition (SD) video camera that was mounted 1.575 m above the seafloor in the center of the sampling frame. Each quadrant covered about 2.8 m^2 (Figure 3.4). From 2017 onwards the survey used a much

higher resolution digital still camera as the primary camera, with a field of view of approximately 2.3 m² (Figure 3.4). Although there were no direct calibrations between this camera and the SD video camera, previous comparisons between the SD video camera and other cameras deployed at the same time showed that a fixed logistic selectivity should be assumed for the ability of the SD video camera to detect different sizes of scallops.

The precision of measurements must be considered when interpreting shell height data from video. Based on tank experiments, Jacobson et al. (2010) estimated measurement error for the video survey with a standard deviation of 6.1 mm. Field measurements are likely less precise than in a tank. For this reason, measurement error was estimated by fitting SMAST shell heights to dredge shell heights from the same year and region that were convolved with a Gaussian kernel with mean 0 and standard deviation σ (NEFSC 2014). The standard deviation that best fit the SMAST shell heights over all years and regions was 11 mm, which was the value used in modeling for this assessment.

The survey coverage of the drop camera varied through the Mid-Atlantic and Georges Bank regions over the years. Therefore, the coverage was standardized to a common footprint for each of these regions. The common footprint for each region was decided based on a balance of maintaining as much survey data as possible and avoiding major extrapolation into unsurveyed areas in certain years. Data that fell outside of the boundaries of the common footprint were removed, and a GAM model, specific to each region, was used to estimate scallop densities in unsurveyed areas within the common footprint for each year. The GAM models were fit to observed mean station densities with a smoother on the covariates latitude, longitude, year, and depth. The models used a Tweedie distribution with the power parameter set at 1.5. The GAM fits were then used to predict mean scallop density on a 2 km by 2 km grid within unsurveyed areas within the common footprint in each year. Mean density, abundance, and associated statistics were then calculated for each region and year based on combined observations and GAM model predictions (Table 3.2). Standard errors used to estimate coefficient of variations for the mean density estimates were calculated assuming a stratified random approach, where different grid sizes were considered separate strata within each region and year.

Habcam towed camera survey

The habitat mapping camera system (Habcam) is an underwater towed digital camera system. The camera(s) take rapid-fire photos of the sea floor (typically 6/sec) as it is towed at speeds between 5-7 knots at roughly 2 m above the bottom. Camera output is transmitted to the vessel via a fiber-optic cable and recorded on a hard disk, along with related metadata. The images were then processed (i.e., debayered from tiffs to pngs, color-corrected, lightmap smoothed, and calibrated stereo rectification) into a human-readable format and manually annotated, targeting species such as sea scallops, fish, crabs, etc.

Four Habcam vehicles have been used. The first, known as “v2”, carried a single camera and was in operation from 2005 to 2012. The other three, known as “v3”, “v4”, and “v5”, have two cameras to allow 3D viewing and obtain precise measurements. The v4 unit also carries a side-scan sonar and a full array of oceanographic sensors (e.g., CTD, chlorophyll,

dissolved oxygen, water spectrometer, etc.) and has been used for broad-scale sea scallop surveys by NEFSC since 2012. The NEFSC Habcam surveys were supplemented with more finescale surveys of specific areas. These surveys were conducted by the Habcam Group/Arnie's Fisheries, and later by the Coonamessett Farm Foundation, Woods Hole Oceanographic Institution, and Coastal Ocean Vision, primarily on the F/V *Kathy Marie*, using the v2, v3 and v5 systems.

NEFSC first tested the Habcam system (v2) in 2009 on Georges Bank in various selected locations. Broad-scale Habcam surveys started in 2011 on Georges Bank using the v2 system, and on both Georges Bank and Mid-Atlantic since 2012 using the v4 system. All NEFSC Habcam surveys were conducted on R/V *Hugh Sharp* until 2024 when R/V *Henry B. Bigelow* was also used.

Because of the large number of images collected, only subsets were examined for sea scallop measurements and counts; typically between 1 in 50 to 1 in 100 photographs were analyzed, corresponding to about one photograph annotated every 25 to 50 meters. These data were interpolated to large scales using zero-inflated generalized additive models followed by ordinary kriging of the residuals (Chang et al. 2017, Figure 3.5). An alternative method, taking stratified means of the transects with accurate stratification, gave similar results (Chang et al. 2017).

The footprint of the broad-scale Habcam survey changed in 2015 to incorporate more areas in the northern edge and Nantucket Lightship in Georges Bank, as well as the deep water portion in the south of the Mid-Atlantic, which were previously considered low/zero-density areas. Moreover, the broad-scale Habcam survey was incomplete in 2019, 2020, 2021, and 2023 in Georges Bank and 2020, 2021, and 2023 in Mid-Atlantic, respectively. To standardize the survey footprint, areas with inconsistent or missing survey coverage were either dropped from the time series (2020 and 2023 for Georges Bank) or estimated using data from adjacent years and the same regression kriging method with an additional year factor. The estimated abundance and biomass from the Habcam survey are in Table 3.3.

Comparison of surveys

The biomass and abundance estimates for the three survey gears are shown in Figure 3.6 over 2003 to 2023. The abundance estimates from Georges Bank show reasonably similar trends. Prior to the first use of the Habcam survey in 2011, Georges Bank abundance was estimated to be between 2 and 6 billion scallops each year by the dredge and drop camera surveys. All surveys estimated time-series peaks in abundance of 8 to 10 billion scallops between 2014 and 2017, with peak abundance occurring in 2014 for Habcam, 2015 for the dredge, and 2017 for the drop camera. After this period all surveys estimated substantial declines in abundance from these peak values to estimates of between 3 and 6 billion scallops per year.

Biomass estimates on Georges Bank were variable through time, which captured much of the variability from the abundance time series and the additional variability introduced by

variations in shell height distributions observed by each of the surveys. Although there were some years of agreement on biomass magnitude among the surveys, there were also several years with considerably different estimates. Both the Habcam and drop camera surveys estimated large peaks in biomass of around 150,000 metric tons between 2016 and 2018, whereas the dredge survey estimated biomass to be 45,000 to 90,000 metric tons during this period. Both the Habcam and drop camera surveys estimated rapid declines in biomass until 2022 from their respective peaks and their estimates closely aligned with the dredge biomass estimates from 2021 to 2023.

The trends of abundance estimates through time from the three surveys in the Mid Atlantic were very similar to each other. Abundance was estimated as 11 to 12 billion scallops by both the dredge and drop camera surveys in 2003 before decline in abundances to between 2 and 3 billion scallops in 2011. The Habcam survey was first used in the Mid Atlantic in 2012 and all three surveys saw increases in abundance from 2012 through to observed high estimated abundances in 2015 of between 10 and 16 billion scallops. After the high 2015 abundances, the survey estimated abundances declined rapidly over the next five years before stabilizing at around 2 billion scallops per year from 2020 to 2023. Biomass estimates in the Mid Atlantic were also reasonably well aligned among the three surveys, and followed the same patterns as the abundance estimates.

Scallops in the Deep Southeast Portion of Nantucket Lightship Closed Area

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The very large 2012 year class in Georges Bank included very dense settlement in the relatively deep water in the southeast portion of Nantucket Lightship Closed Area, where previously there had been very few scallops (Table 3.4). Scallops in this area grew very slowly, and were nicknamed “Peter Pan” scallops, because they did not want to grow up.

In 2014, many of these scallops were below the 40 mm threshold for inclusion in the survey gear abundance estimates, and were only partially selected to the drop camera survey due to the lower-resolution standard definition video camera used in this survey until 2017. The peak abundance in 2015 was about 11 billion scallops, according to the Habcam survey, an extraordinary number that exceeds the number of scallops in the whole resource for some years. Maximum biomass was in 2017 at about 63000 mt meats. Estimates for scallops in this area have declined across all surveys from these peaks, with all surveys estimating less than 50 million scallops above 40 mm shell height in this area in 2023.

Gulf of Maine surveys

Scallop surveys have been conducted in the Gulf of Maine federal waters using the S Mast drop camera and the Maine Department of Marine Resources (DMR) unlined survey dredge since 2009. The data from these surveys are not currently used in the CASA or SYM models, but they are detailed in the Gulf of Maine working paper.

Tables

Table 3.1. Dredge survey indices. Relative Abundance (Rel.Ab) and Relative Biomass (Rel.Bms) are in per tow units, whereas Absolute Abundance (Abs.Ab) and Absolute Biomass (Abs.Bms) are expanded using assumed dredge efficiency. Prop.Pos is the proportion of tows which caught at least one scallop.

(a) Georges Bank without Deep Southeast Nantucket Lightship (DSENLS) scallops.

Year	Rel.Ab num/tow	CV	Rel.Bms kg/tow	CV	Ntows	Prop. Pos	Mean Wt	Abs.Ab millions	Ab.Bms mt
1979	87.4	0.41	1.758	0.34	108	0.89	20.1	1269	25522
1980	75.8	0.24	0.844	0.16	118	0.81	11.1	1031	11469
1981	61.2	0.13	1.065	0.13	82	0.83	17.4	753	13094
1982	132.9	0.46	1.05	0.32	118	0.83	7.9	2076	16407
1983	61.2	0.22	0.783	0.21	126	0.88	12.8	890	11392
1984	39.3	0.11	0.55	0.1	128	0.85	14	536	7513
1985	61.8	0.15	0.713	0.16	154	0.9	11.5	830	9573
1986	116.8	0.13	0.795	0.1	153	0.9	6.8	1445	9832
1987	120.1	0.17	1.055	0.16	170	0.86	8.8	1619	14223
1988	98.7	0.16	0.881	0.14	175	0.8	8.9	1289	11502
1989	63.6	0.11	0.525	0.08	120	0.78	8.3	806	6655
1990	184.1	0.24	1.222	0.22	175	0.81	6.6	2415	16028
1991	257.9	0.37	1.384	0.25	176	0.89	5.4	3678	19737
1992	232	0.44	1.933	0.43	171	0.89	8.3	3300	27492
1993	61.8	0.24	0.484	0.16	164	0.87	7.8	753	5892
1994	46.7	0.2	0.472	0.16	177	0.84	10.1	561	5672
1995	111.8	0.2	0.788	0.16	176	0.88	7	1637	11533
1996	133.6	0.2	1.624	0.19	171	0.9	12.2	1855	22555
1997	89.4	0.15	1.721	0.17	190	0.88	19.3	1292	24884
1998	283	0.26	3.828	0.32	195	0.87	13.5	3646	49304
1999	193.5	0.15	3.361	0.16	173	0.98	17.4	2663	46251

2000	766.7	0.29	7.364	0.22	164	0.91	9.6	9996	96008
2001	408.9	0.13	6.532	0.13	208	0.95	16	5560	88826
2002	334.5	0.14	7.043	0.14	214	0.93	21.1	4498	94727
2003	277.9	0.12	7.051	0.13	207	0.94	25.4	3839	97407
2004	291.5	0.11	8.498	0.12	218	0.94	29.2	3959	115432
2005	265.6	0.12	6.9	0.09	343	0.95	26	3888	101024
2006	221.3	0.13	6.439	0.13	236	0.94	29.1	3258	94792
2007	224.8	0.1	4.953	0.07	363	0.97	22	3453	76092
2008	321.8	0.1	6.542	0.08	239	0.97	20.3	4805	97676
2009	362.7	0.15	5.831	0.11	214	0.97	16.1	5497	88380
2010	413.1	0.21	7.468	0.09	268	0.97	18.1	6407	115838
2011	279.4	0.12	6.781	0.08	225	0.96	24.3	3946	95787
2012	225.3	0.13	4.97	0.08	224	0.97	22.1	3488	76943
2013	336.5	0.23	4.599	0.14	213	0.94	13.7	4416	60352
2014	519.8	0.47	3.859	0.18	124	0.9	7.4	8439	62644
2015	645.2	0.28	5.54	0.24	178	0.94	8.5	8975	75958
2016	195.1	0.14	3.047	0.16	310	0.9	15.8	2612	41291
2017	244.0	0.17	4.954	0.16	280	0.88	20.3	3143	63908
2018	262.6	0.18	7.01	0.16	372	0.87	27.9	3518	98002
2019	313.1	0.06	5.616	0.05	371	0.91	17.9	4290	76940
2020	297.4	0.07	5.131	0.07	328	0.90	13.8	4193	72346
2021	172.7	0.08	3.418	0.08	280	0.89	19.6	2977	58475
2022	404.8	0.06	3.579	0.07	324	0.83	8.7	7015	61231
2023	293.2	0.08	4.457	0.07	322	0.81	15.2	4881	73991

Table 3.1(b) Mid-Atlantic

Year	Rel.Ab num/tow	CV	Rel.Bms kg/tow	CV	Ntows	Prop. Pos	Mean Wt	Abs.Ab millions	AbsBms mt
1979	34.7	0.1	0.609	0.1	166	0.92	17.6	550	9661

1980	42.8	0.12	0.518	0.08	167	0.94	12.1	679	8223
1981	32.1	0.16	0.403	0.13	167	0.91	12.6	509	6391
1982	33.5	0.11	0.435	0.08	185	0.91	13	532	6898
1983	32.3	0.1	0.404	0.08	193	0.89	12.5	512	6416
1984	32.2	0.11	0.383	0.09	204	0.91	11.9	510	6083
1985	74.1	0.12	0.595	0.09	201	0.94	8	1177	9438
1986	129.6	0.09	1.028	0.08	226	0.93	7.9	2056	16310
1987	131.9	0.08	0.917	0.07	226	0.93	6.9	2093	14548
1988	147.8	0.1	1.423	0.08	227	0.91	9.6	2345	22590
1989	172.8	0.09	1.207	0.07	244	0.93	7	2742	19159
1990	215.2	0.22	1.335	0.18	216	0.89	6.2	3415	21193
1991	81	0.1	0.775	0.1	228	0.92	9.6	1285	12292
1992	43.5	0.11	0.441	0.07	229	0.87	10.1	690	6991
1993	135.6	0.1	0.615	0.08	214	0.96	4.5	2152	9768
1994	145.1	0.13	0.866	0.09	227	0.94	6	2302	13737
1995	173.4	0.13	1.218	0.11	227	0.96	7	2751	19325
1996	58.8	0.08	0.606	0.07	211	0.89	10.3	933	9612
1997	43.2	0.13	0.43	0.06	225	0.93	9.9	686	6824
1998	168.4	0.15	0.828	0.12	215	0.92	4.9	2672	13137
1999	238.3	0.24	1.734	0.2	226	0.92	7.3	3782	27524
2000	292.1	0.14	3.028	0.13	229	0.88	10.4	4636	48058
2001	308.4	0.11	3.419	0.12	227	0.9	11.1	4894	54257
2002	284	0.1	3.578	0.11	206	0.89	12.6	4508	56791
2003	654.5	0.16	5.766	0.1	201	0.9	8.8	10387	91512
2004	471	0.12	5.061	0.08	248	0.89	10.7	7475	80322
2005	344.6	0.08	5.406	0.07	278	0.94	15.7	5469	85792
2006	386.6	0.09	6.138	0.07	302	0.95	15.9	6136	97420
2007	314.6	0.06	5.487	0.06	304	0.94	17.4	4994	87078
2008	373.7	0.09	5.549	0.08	259	0.97	14.8	5932	88061
2009	370.5	0.12	6.134	0.1	196	0.92	16.6	5880	97351

2010	250.3	0.08	4.625	0.07	281	0.94	18.5	3973	73407
2011	172.7	0.1	3.482	0.1	298	0.96	20.2	2740	55264
2012	260.2	0.12	2.728	0.06	269	0.94	10.5	4130	43295
2013	256.1	0.1	3.201	0.08	309	0.98	12.5	4065	50801
2014	255.9	0.17	3.664	0.12	445	0.9	14.3	4061	58154
2015	624.1	0.12	3.923	0.09	495	0.95	6.3	9904	62255
2016	352.8	0.16	3.999	0.13	434	0.92	11.3	6782	76885
2017	316.6	0.15	4.740	0.14	423	0.92	15.0	6087	91131
2018	198.2	0.14	3.885	0.13	427	0.89	19.6	3821	74900
2019	127.8	0.12	3.098	0.11	436	0.83	24.2	2397	58104
2020	60.6	0.08	1.479	0.07	431	0.81	24.4	1168	28503
2021	59.2	0.11	1.108	0.08	432	0.84	18.7	1140	21352
2022	86.5	0.09	1.249	0.08	432	0.76	14.4	1668	24077
2023	43.8	0.08	0.806	0.09	432	0.76	18.4	845	15540

Table 3.1(c) Georges Bank and Mid-Atlantic combined, without DSENLS scallops.

Year	Rel.Ab num/tow	CV	Rel.Bms kg/tow	CV	Ntows	Prop. Pos	Mean Wt	Abs.Ab millions	Ab.Bms mt
1979	57.6	0.26	1.108	0.23	274	0.91	19.3	1819	35183
1980	57.2	0.14	0.66	0.09	285	0.89	11.5	1710	19692
1981	44.7	0.1	0.69	0.09	249	0.88	15.4	1261	19486
1982	76.7	0.33	0.702	0.19	303	0.88	8.9	2608	23305
1983	44.8	0.13	0.569	0.12	319	0.88	12.7	1402	17808
1984	35.3	0.08	0.456	0.07	332	0.89	13	1047	13597
1985	68.8	0.09	0.646	0.09	355	0.92	9.5	2007	19011
1986	124	0.07	0.927	0.06	379	0.92	7.5	3501	26143
1987	126.8	0.09	0.977	0.08	396	0.9	7.8	3712	28771
1988	126.5	0.08	1.188	0.07	402	0.86	9.4	3634	34092
1989	125.3	0.08	0.911	0.06	364	0.88	7.3	3548	25814
1990	201.7	0.16	1.286	0.14	391	0.85	6.4	5830	37221
1991	157.8	0.27	1.039	0.15	404	0.91	6.5	4963	32029

1992	125.4	0.35	1.089	0.33	400	0.88	8.6	3990	34483
1993	103.6	0.1	0.558	0.08	378	0.92	5.4	2905	15661
1994	102.4	0.11	0.695	0.08	404	0.9	6.8	2864	19408
1995	146.6	0.11	1.031	0.09	403	0.92	7.0	4388	30858
1996	91.3	0.13	1.048	0.13	382	0.9	11.5	2788	32167
1997	63.3	0.11	0.991	0.13	415	0.91	16.0	1978	31708
1998	218.2	0.17	2.13	0.26	410	0.9	9.9	6318	62441
1999	218.8	0.16	2.441	0.13	399	0.95	11.4	6445	73775
2000	498.2	0.2	4.911	0.15	393	0.89	9.8	14633	144066
2001	352	0.09	4.771	0.1	435	0.93	13.7	10454	143083
2002	305.9	0.09	5.083	0.1	420	0.91	16.8	9006	151518
2003	490.9	0.12	6.324	0.09	408	0.92	13.3	14226	188919
2004	393	0.09	6.554	0.08	466	0.91	17.1	11434	195754
2005	310.3	0.07	6.055	0.06	621	0.95	20.0	9357	186816
2006	314.8	0.08	6.269	0.07	538	0.95	20.5	9394	192211
2007	275.6	0.06	5.255	0.05	667	0.95	19.3	8447	163170
2008	351.2	0.07	5.98	0.06	498	0.97	17.3	10737	185737
2009	367.1	0.1	6.002	0.07	410	0.95	16.3	11377	185731
2010	321	0.13	5.86	0.06	549	0.95	18.2	10380	189244
2011	219	0.08	4.915	0.06	523	0.96	22.6	6687	151051
2012	245	0.09	3.702	0.06	493	0.96	15.8	7618	120238
2013	291	0.12	3.808	0.1	659	0.96	13.1	8481	111154
2014	370.5	0.3	3.749	0.12	569	0.9	9.7	12500	120798
2015	633.3	0.17	4.627	0.17	673	0.95	7.3	18879	138213
2016	284.1	0.1	3.584	0.1	744	0.91	12.6	9394	118176
2017	285	0.12	4.833	0.11	703	0.9	16.8	9230	155039
2018	226.3	0.13	5.246	0.13	799	0.88	1205	7339	172902
2019	208.5	0.06	4.195	0.05	807	0.87	20.2	6687	135044
2020	163.7	0.07	3.07	0.07	759	0.85	18.8	5361	100849
2021	108.6	0.08	2.114	0.08	712	0.86	19.4	4117	79827
2022	225.1	0.06	2.264	0.07	756	0.79	9.8	8683	85308

2023	152.4	0.09	2.396	0.07	754	0.78	15.6	5726	89531
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Table 3.2. SMAST drop camera survey estimates combined from observations and GAM modelling, excluding DSENLIS scallops. Note that high resolution digital cameras were used from 2017-2023 whereas 2003-15 used standard definition video cameras. All estimates correspond to scallops greater than or equal to 40 mm shell height only.

Year	Observed Density (m ⁻²)	Observed CV	Observed Area (km ²)	Observed Abundance (millions)	Projected Abundance (millions)	Projected Area (km ²)	Total Abundance (millions)	Total biomass (metric tons)
Georges Bank Open								
2003	0.086	0.09	17163.2	1482	197	1728.7	1679	19608
2004	0.067	0.13	16391.7	1080	281	2500.2	1361	18074
2005	0.063	0.11	16576.7	1048	177	2315.2	1225	19477
2006	0.067	0.14	16711.3	1118	124	2180.6	1243	21559
2007	0.103	0.16	16129.2	1657	644	2762.7	2301	29890
2008	0.078	0.11	17514.1	1360	150	1377.8	1511	21512
2009	0.126	0.16	16692.5	2105	102	2199.4	2207	25797
2010	0.043	0.17	16623.0	707	87	2268.9	794	15356
2011	0.076	0.20	17225.0	1317	71	1666.9	1388	27880
2012	0.086	0.15	16684.8	1429	153	2207.1	1583	32122
2014	0.079	0.12	17291.5	1373	319	1600.4	1692	20707
2015	0.100	0.14	18891.9	1890	0	0.0	1890	17286
2017	0.109	0.10	14385.0	1567	243	4506.9	1810	23385
2018	0.077	0.10	4267.7	329	938	14624.2	1267	24694
2019	0.060	0.16	6034.9	365	968	12857.0	1332	27014
2020	0.254	0.11	5456.1	1388	1120	13435.8	2508	45276
2021	0.195	0.10	11699.4	2286	103	7192.5	2388	43855
2022	0.150	0.10	11699.4	1757	193	7192.5	1950	28397
2023	0.257	0.47	11907.8	3065	214	6984.1	3278	36528
Georges Bank Closed								
2003	0.258	0.12	10495.5	2710	118	3489.5	2828	78680
2004	0.229	0.10	10297.7	2362	81	3687.3	2443	65975

2005	0.198	0.11	9955.2	1968	79	4029.8	2047	51031
2006	0.211	0.07	10679.2	2256	97	3305.8	2353	78338
2007	0.160	0.08	10804.7	1729	112	3180.3	1841	59276
2008	0.133	0.11	10681.3	1417	163	3303.7	1580	34147
2009	0.228	0.10	9837.6	2243	135	4147.4	2378	54431
2010	0.163	0.11	10132.8	1647	61	3852.2	1708	48081
2011	0.148	0.16	10773.3	1600	106	3211.7	1706	38411
2012	0.173	0.18	10418.3	1806	140	3566.7	1946	34677
2014	0.290	0.24	10957.3	3174	189	3027.7	3364	33638
2015	0.408	0.42	9675.3	3947	468	4309.7	4415	39323
2017	0.540	0.26	11691.7	6309	91	2293.3	6401	132389
2018	0.928	0.17	5078.0	4712	1049	8907.0	5762	142603
2019	0.308	0.28	5679.9	1752	1378	8305.1	3131	78926
2020	0.328	0.13	5170.6	1695	840	8814.4	2535	50151
2021	0.143	0.13	12694.9	1818	24	1290.1	1841	29758
2022	0.171	0.21	12671.8	2164	28	1313.2	2192	22854
2023	0.215	0.19	12478.8	2679	6	1506.2	2685	48083

Mid Atlantic

2003	0.463	0.16	23367.9	10814	366	4756.4	11180	105169
2004	0.236	0.10	24787.9	5841	204	3336.4	6045	79421
2005	0.214	0.13	24973.1	5334	142	3151.2	5476	80459
2006	0.205	0.10	25309.0	5182	135	2815.4	5317	63088
2007	0.202	0.10	25395.4	5131	169	2728.9	5301	83608
2008	0.205	0.10	25189.2	5173	173	2935.1	5347	84850
2009	0.152	0.08	23648.2	3598	510	4476.1	4109	66065
2010	0.067	0.09	21452.8	1451	620	6671.5	2071	39332
2011	0.049	0.08	24165.6	1185	146	3958.8	1331	33861
2012	0.116	0.09	27226.6	3171	39	897.8	3210	31907
2014	0.133	0.07	27350.1	3638	44	774.3	3682	54516
2015	0.457	0.19	23800.1	10870	584	4324.3	11454	102525
2017	0.306	0.26	25474.8	7797	106	2649.6	7903	149269
2019	0.118	0.09	23784.7	2815	226	4339.7	3041	63027

Combined								
2003	0.294	0.12	51026.6	15006	681	9974.6	15687	203457
2004	0.181	0.07	51477.3	9283	566	9523.9	9849	163470
2005	0.162	0.09	51505.0	8350	398	9496.2	8748	150967
2006	0.162	0.07	52699.5	8556	356	8301.8	8913	162985
2007	0.169	0.07	52329.3	8517	925	8671.9	9443	172774
2008	0.149	0.07	53384.6	7950	486	7616.6	8438	140509
2009	0.158	0.06	50178.3	7946	747	10822.9	8694	146293
2010	0.079	0.07	48208.6	3805	768	12792.6	4573	102770
2011	0.078	0.09	52163.9	4102	323	8837.4	4425	100153
2012	0.118	0.08	54329.7	6406	332	6671.6	6739	98706
2014	0.147	0.10	55598.9	8185	552	5402.4	8738	108861
2015	0.319	0.16	52367.3	16707	1052	8634.0	17759	159134
2017	0.304	0.17	51551.5	15673	440	9449.8	16114	305042
2019	0.139	0.11	35499.5	4932	2572	25501.8	7504	168967

Table 3.3. Habcam survey estimates, excluding DSENLs scallops. The 2011 survey used the single camera “v2” system, and the 2012-2023 ones used the stereo camera “v4” system.

Year	Number millions	CV	Biomass mt	CV
Georges Bank				
2011	3950	0.19	102676	0.16
2012	4420	0.12	90014	0.07
2013	3524	0.05	47351	0.13
2014	10431	0.09	73173	0.17
2015	8756	0.05	115188	0.06
2016	9751	0.05	148920	0.10
2017	6304	0.05	130355	0.04
2018	4801	0.06	105640	0.05
2019	3675	0.05	45998	0.05
2020	NA		NA	
2021	3548	0.07	61052	0.06
2022	3713	0.07	46856	0.06
2023	NA		NA	

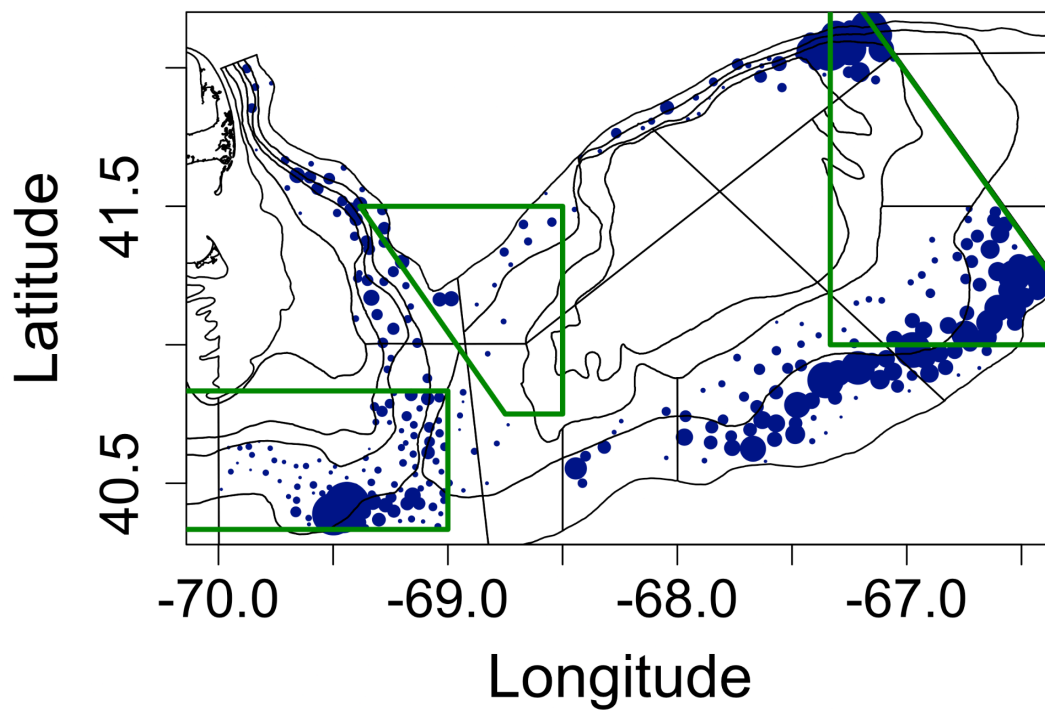
Mid Atlantic

2012	4666	0.05	49186	0.11
2013	4218	0.15	60816	0.05
2014	5604	0.06	88365	0.15
2015	15755	0.16	114481	0.10
2016	11212	0.29	121781	0.20
2017	7556	0.11	111677	0.07
2018	3693	0.08	89032	0.05
2019	3341	0.11	81715	0.08
2020	1493	0.19	47599	0.25
2021	1396	0.12	24958	0.11
2022	1493	0.05	20393	0.07
2023	1185	0.08	17934	0.31
Combined				
2012	9086	0.08	139200	0.08
2013	7741	0.10	108167	0.09
2014	16035	0.08	161538	0.16
2015	24510	0.12	229669	0.08
2016	20963	0.17	270701	0.14
2017	13860	0.07	242032	0.05
2018	8494	0.07	194672	0.04
2019	7016	0.06	127713	0.06
2020	NA		NA	
2021	4943	0.08	86010	0.08
2022	5205	0.07	67250	0.06
2023	NA		NA	

Table 3.4. Survey estimates of DSENLS scallops. “MeanMW”, “BmsMw”, are mean meat weight and biomass in meat weight, respectively, all from Habcam data. Survey numbers are restricted to those greater than 40 mm shell height; in 2014, many of the scallops were smaller than this size.

Year	Dredge mill.	CV	DropCam mill.	CV	Habcam mill.	CV	Mean MW g	Bms MW mt
2014	N/S	N/S	4708	0.29	201	0.02	1.8	353
2015	N/S	N/S	4012	0.35	11083	0.11	4.4	48567
2016	5598	0.24	3346	0.26	10664	0.04	4.2	44803
2017	3152	0.26	11043	0.31	8251	0.03	7.6	63054
2018	1248	0.09	5031	0.19	4070	0.09	14.3	58355
2019	1176	0.1	4211	0.19	3829	0.05	12.0	46060
2020	3613	0.21	2687	0.19	2441	0.05	11.5	28001
2021	802	0.1	1925	0.42	1141	0.04	13.4	15317
2022	162	0.13	172	0.64	709	0.05	15.6	11090
2023	22	0.25	42	1.00	17	0.05	10.8	184

Figures



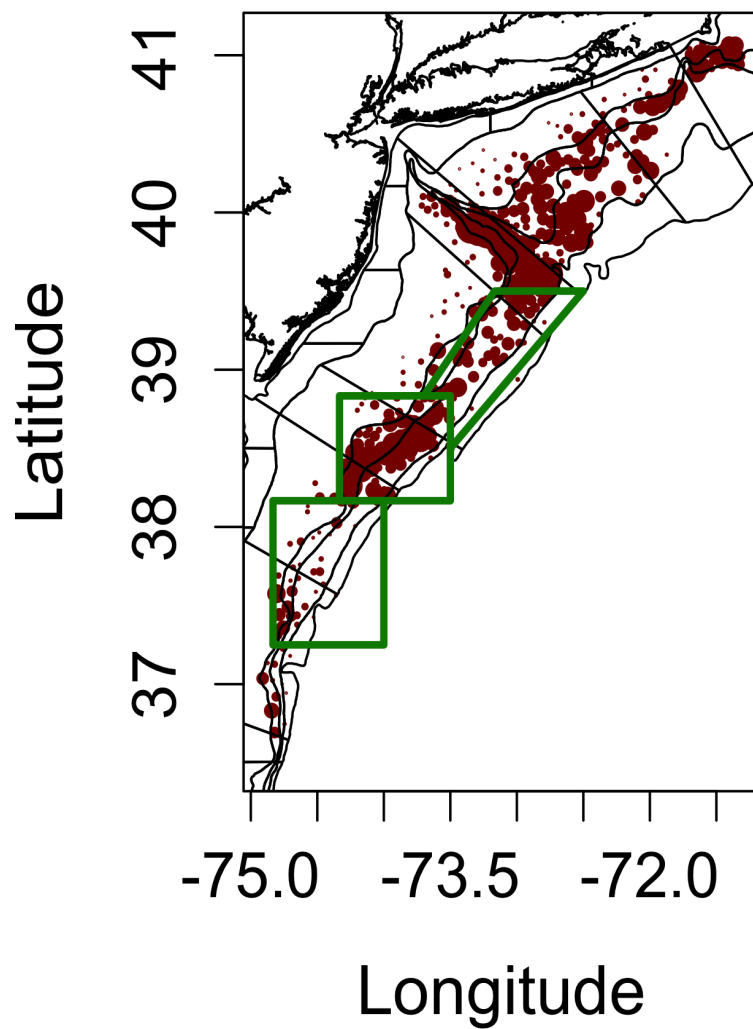
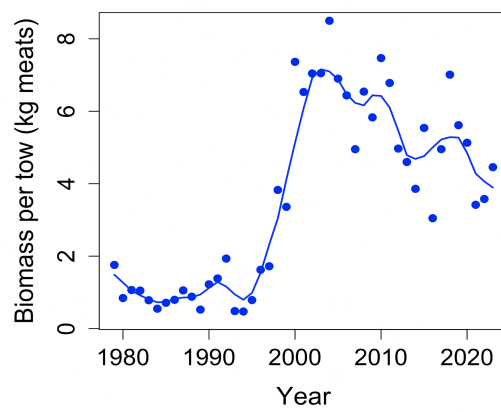
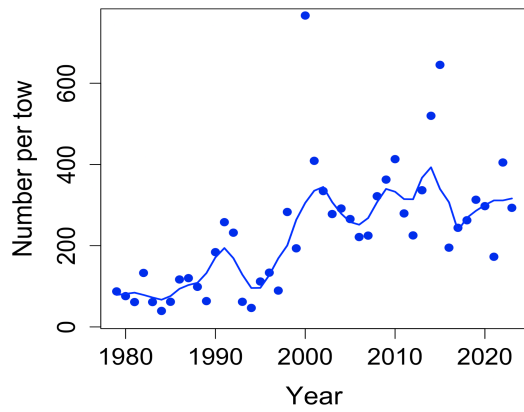


Figure 3.1. Dredge survey charts, showing 2022 catches, for the Georges Bank (above) and Mid-Atlantic (below) regions. Rotational and closed areas are shown in green, and the survey strata are shown in black.

(a) Georges Bank



(b) Mid-Atlantic

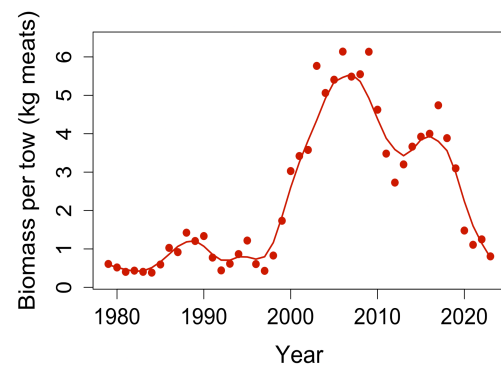
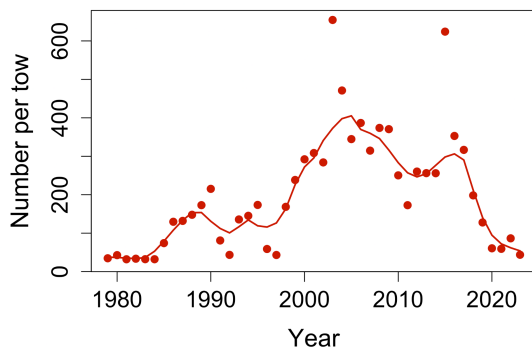


Figure 3.2. Dredge time series for numbers (left) and biomass (right) (dots), including lowess smoothers (lines) for (a) Georges Bank and (b) Mid-Atlantic.

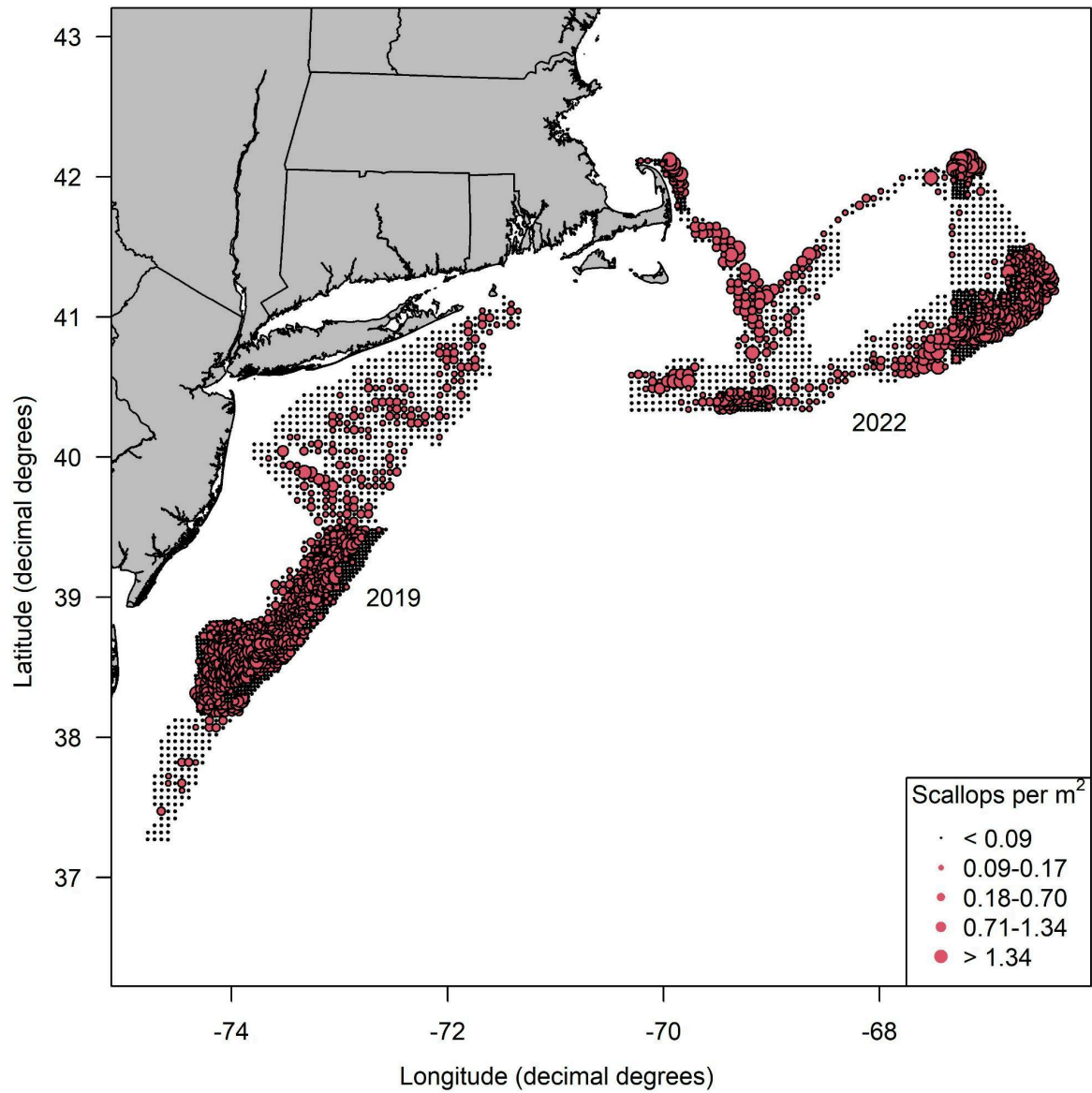


Figure 3.3. An example of drop camera survey coverage and corresponding scallop densities, with the Mid Atlantic region coverage and densities on the left coming from 2019 and the Georges Bank coverage and densities in the northeast of the map coming from 2022.

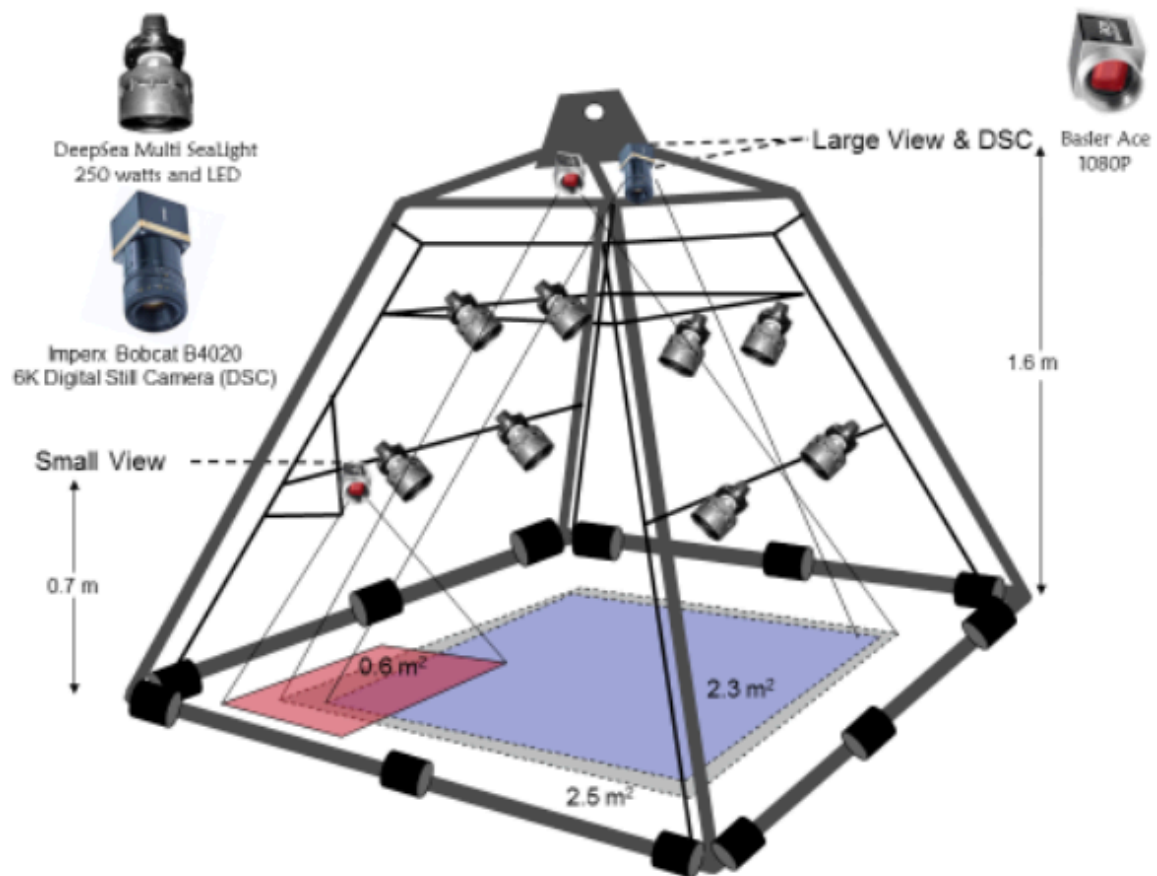


Figure 3.4. The drop camera pyramid setup that has been used since 2017, with a variety of cameras including the “large cam” used prior to 2017.

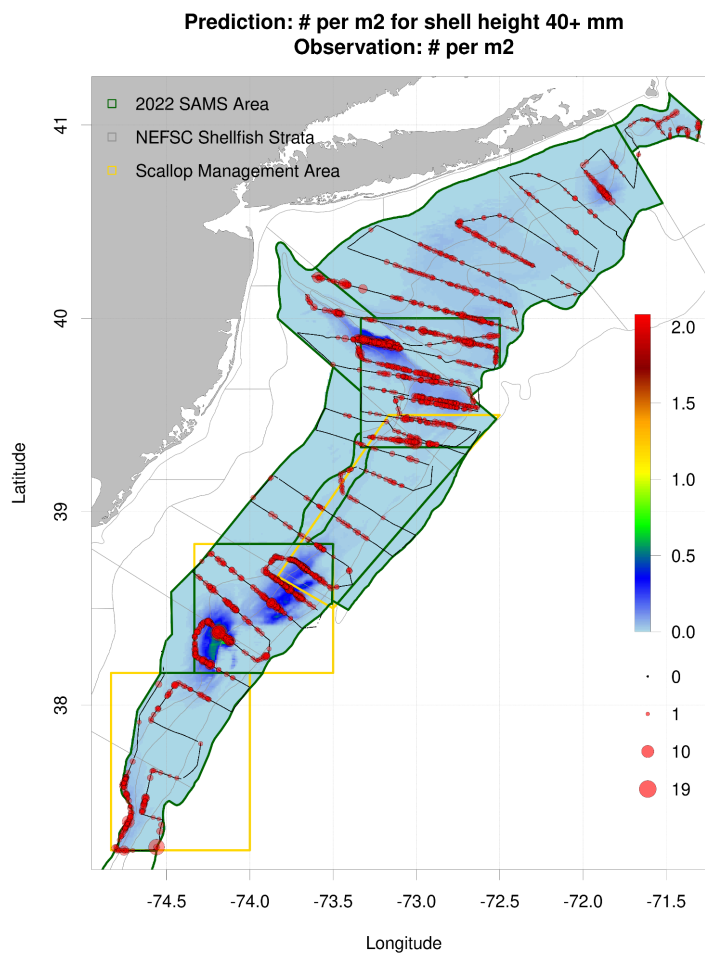
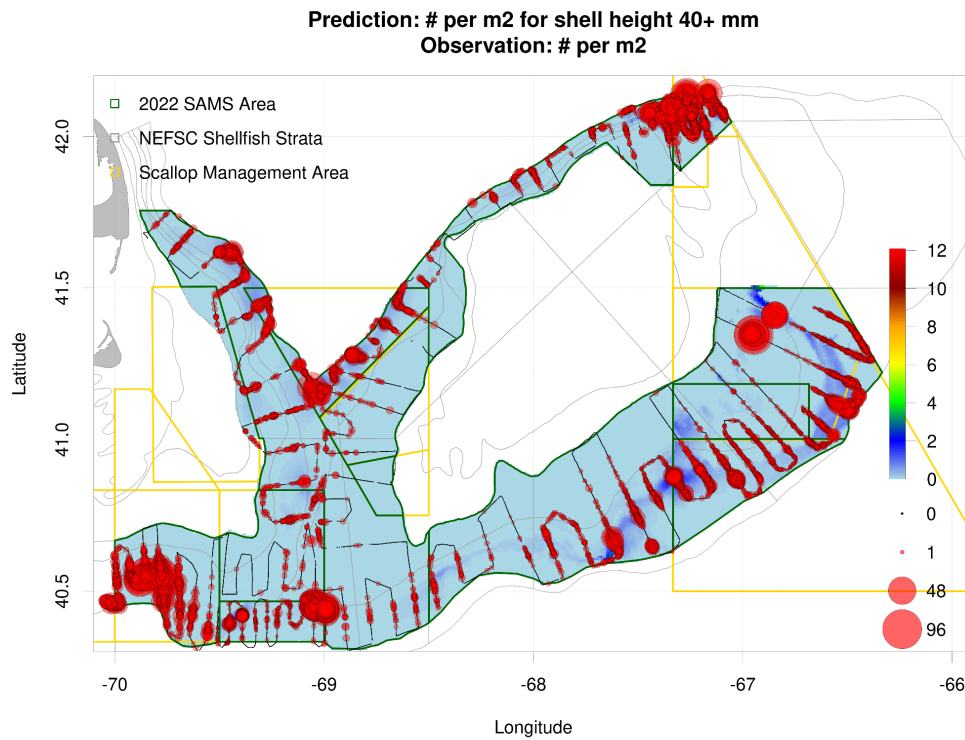


Figure 3.5. Habcam survey charts, showing survey track conducted by NEFSC and Coonamessett Farm Foundation, and geostatistical model estimated densities for 2022.

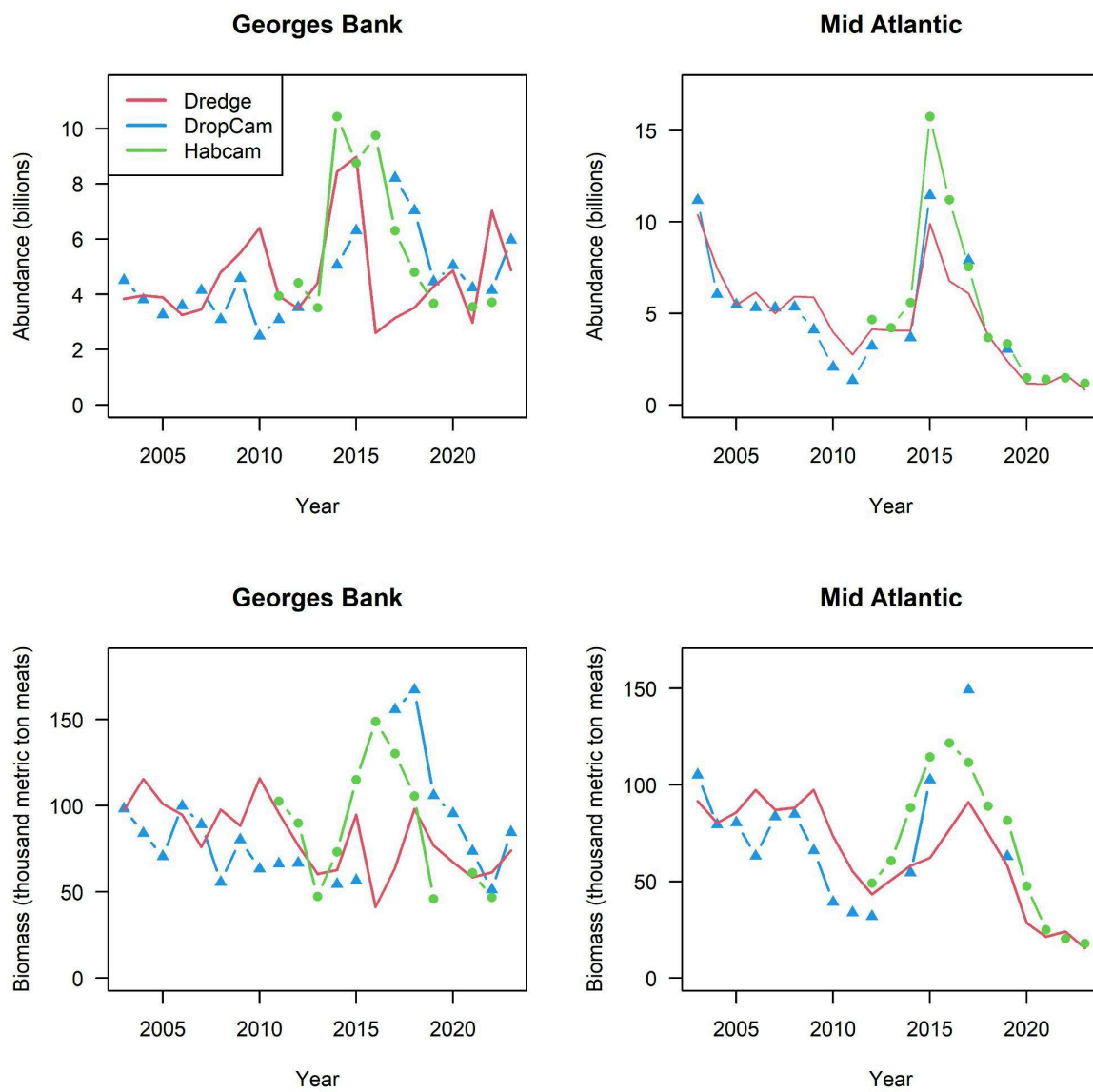


Figure 3.6. Comparison of survey estimates, 2003-2023, for Georges Bank (left), and Mid-Atlantic (right). Numbers are on the top row, and biomass on the bottom row.